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A meta-analysis suggests that TMS targeting the hippocampal network selectively improves episodic memory

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This meta-analysis provides a **fundamental** synthesis of evidence demonstrating that transcranial magnetic stimulation targeting the hippocampal-cortical network reliably enhances episodic memory performance across diverse study designs. The evidence is **convincing**, with rigorous methodology and consistent effects observed despite modest sample sizes and some heterogeneity in stimulation approaches. The work highlights the specificity of memory improvements to hippocampal-dependent memories and identifies key methodological factors-such as individualized targeting-that influence efficacy. Overall, this study offers a timely and integrative framework that will inform both basic memory research and the design of future clinical trials for cognitive enhancement.

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Abstract

Episodic memory is critically dependent on the hippocampal network and is frequently impaired in many clinical disorders. Recent findings highlight Hippocampal Indirectly Targeted Stimulation (HITS) as a promising, network-guided noninvasive transcranial magnetic stimulation (TMS) procedure to enhance episodic memory performance. Here, we report the first comprehensive meta-analysis of HITS effects on episodic memory, encompassing both healthy individuals and clinical populations. HITS using parieto-occipital network targets robustly improved episodic memory, with effects selective for episodic memory versus other non-memory cognitive domains. Efficacy was significantly greater when memory performance was assessed using memory tasks sensitive to recollection, which is strongly linked to hippocampal network function, compared to recognition or other types of episodic memory tasks. Efficacy was also significantly greater when HITS was delivered before the memory tasks were administered versus in the period between study and test phases of tasks. No serious adverse events were reported. These findings establish HITS as a robust approach for episodic memory enhancement, suggesting potential for clinical translation in memory disorders. Selectivity of effects for episodic memory generally and for recollection-format tests in particular indicates cognitive and mechanistic specificity, supporting the potential for targeted and selective neuromodulation of hippocampal networks and their associated functions.

Impact statement

Meta-analysis indicates that network-targeted noninvasive brain stimulation consistently enhances memory function supported by the hippocampal network, thus providing robust evidence that specific memory abilities rely on specific modifiable brain networks.

Introduction

Memory for episodes of experience (episodic memory) depends critically on the hippocampus and interactions of hippocampus with a set of brain regions that comprise a hippocampal-centered network (1–8). Episodic memory impairments due to a variety of neurological and psychiatric conditions have been associated with disrupted functional connectivity of hippocampal-centered networks, as measured via methods such as functional MRI (3, 4, 6, 9–14). Thus, to the extent that memory impairments result from disrupted network function, modulation of hippocampal network functional connectivity is a potential therapeutic strategy for episodic memory rescue.

Transcranial magnetic stimulation (TMS) induces electrical fields in the human brain with sufficient intensity to trigger focal action potential firing in neocortex (15). Repetitive TMS (rTMS) protocols influence neuroplasticity of distributed networks of the stimulated location, which can influence functional connectivity (16–21). This motivates the hypothesis that rTMS of hippocampal network locations could affect network functional connectivity and thereby influence episodic memory. Indeed, Wang et al. (8) applied multi-day rTMS to a parietal-cortex location identified via its high fMRI connectivity with the hippocampus. They reported that stimulation increased fMRI connectivity of the indirectly targeted portion of hippocampus with its network and improved episodic memory performance. We refer to this stimulation method as “Hippocampal Indirectly Targeted Stimulation” (HITS), as the overall goal is to use noninvasive rTMS to indirectly affect hippocampal network interactions. A variety of findings suggest that HITS modulates activity of brain networks defined relative to the hippocampal indirect target (22, 23). It is possible that HITS could be used to enhance memory function, both in healthy individuals and those with clinical memory impairments resulting from abnormal hippocampal network connectivity (24–26). However, the variety of neuromodulation protocols used by various research groups and variability in reported effects of HITS on memory (22, 27–29) motivate systematic and quantitative evaluation of this potential.

Here, we report results from a systematic review and meta-analysis of studies that investigated the effects of HITS on episodic memory. We reviewed studies in which rTMS was applied to parieto-occipital locations of the hippocampal network (Figure 1 [↗](#)) and that assessed effects of stimulation on objective tests of episodic memory, conducted in healthy young and older adults and in clinical populations with memory impairments. Our main analysis goal was to determine whether HITS affects episodic memory reliably across studies, despite variation in study designs. We also tested whether the effects of HITS were greater on episodic memory than on other cognitive functions measured in the same studies, and whether study design factors modulated the effectiveness of HITS. We focused on hypothesis-driven design factors, including task format, targeting method, and study population, as well as eight factors identified *post hoc* as varying across studies (Table S1 [↗](#)). Notably, the reviewed studies included a range of design factors, including whether the episodic-memory tasks used recollection versus recognition formats, whether subjects were healthy versus memory impaired, whether the experiments followed acute/basic-science versus chronic/clinical-intervention design models, and other various aspects of trial design. Our analysis goal was to determine whether the effects of HITS on performance were statistically robust irrespective of these factors and whether the experiment designs significantly modulated the effects of HITS.

Results

Sample characteristics

Our search identified 47 studies meeting inclusion criteria, 38 of which provided sufficient data to support meta-analysis (Table 1 [↗](#)). These 38 studies included $N = 1,009$ subjects and reported 253 statistical comparisons (i.e., “effects”) of the influence of HITS on performance of episodic-memory (140 effects) and non-memory tasks (113 effects), which we transformed into normalized effects (Hedges’ g). The mean sample size was $n = 23.2$ for episodic-memory effects (range = 4 – 68) and $n = 23.3$ for non-memory effects (range = 8 – 58) (Figure S1 [↗](#)). By definition, stimulation was applied

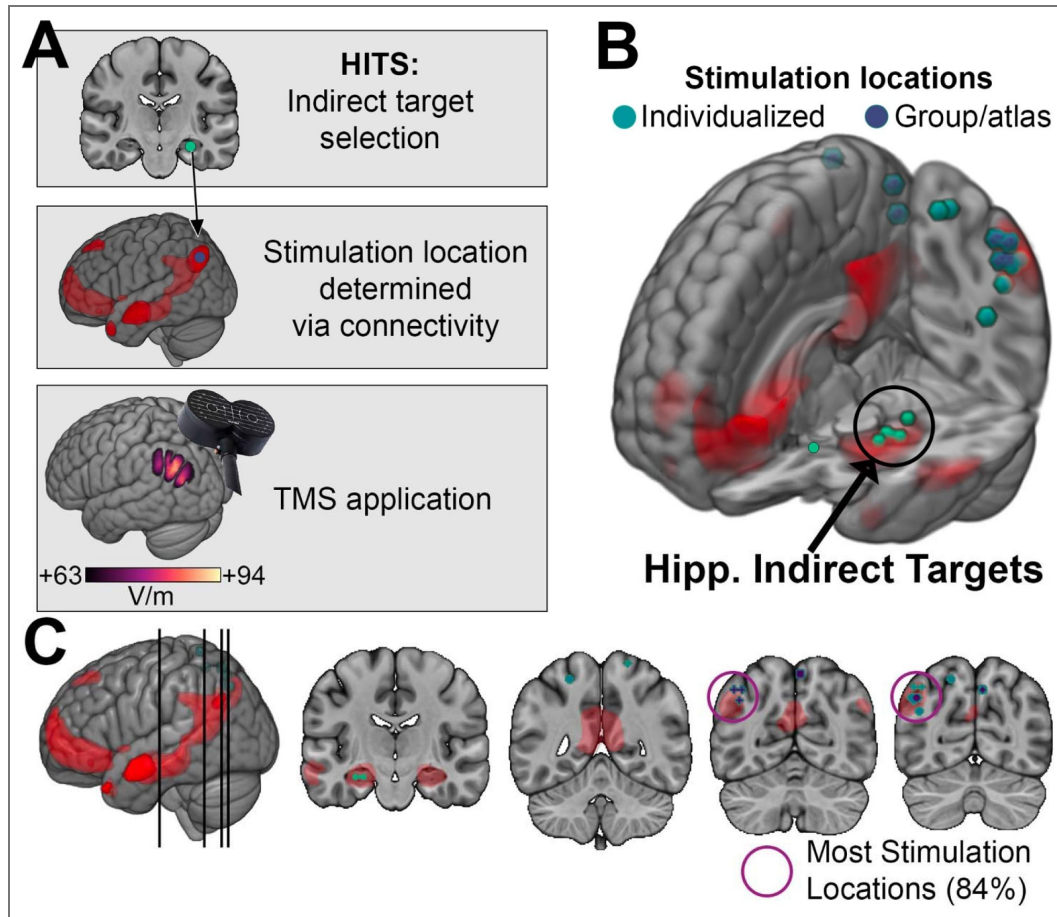


Figure 1. Targets and methods used in HITS experiments.

(A) The general logic of HITS experiments, whereby hippocampal indirect targets are used to define connectivity networks from which stimulation-accessible locations are selected and stimulated with TMS. Red coloration indicates an example group-level fMRI connectivity network of the hippocampal indirect target, from which a stimulation location is selected. A representative electrical field (60) induced by TMS of this location at a typical intensity (estimate of 100% MT) is shown. The field is thresholded at 66% of its maximum intensity, as stimulation of lower intensity should have negligible effects on neuronal activity (all analyzed studies used stimulation intensity of ~70% MT or above). (B) Hippocampal indirect targets and neocortical stimulation locations in the analyzed studies are shown overlaid on a template brain. An example resting-state fMRI connectivity network of the hippocampal indirect targets is displayed in red, as in (A). Stimulation locations are colorized separately for studies that used individualized targeting versus those that used group/atlas-based targets. For studies using individualized targeting, the average (centroid) location of the targets for all subjects in the study is shown. The hippocampal indirect targets are shown for those studies using individualized targeting. Note that not all stimulation locations fall within the highlighted red hippocampal network, as that specific network is shown for illustrative purposes to highlight proximity of all stimulation targets to a typical group-defined hippocampal network. (C) Coronal slices for the indicated positions show the hippocampal indirect targets and stimulation locations in greater detail. K-means clustering indicated that the majority of stimulation locations (84%) comprised a cluster within left parietal cortex, as indicated.

to locations of the hippocampal network that were predominantly in left lateral parietal cortex (84% of studies) or in precuneus or other parieto-occipital neocortex locations generally considered as part of the hippocampal network (all remaining studies) (Figure 1 [↗](#)).

HITS improved episodic memory performance

A meta-regression model with a random effect by study was used to evaluate the overall effect of HITS on episodic memory performance for the 140 effects derived from episodic-memory tasks. The overall effect was positive and highly significant, indicating that HITS improved episodic memory (Hedges' $g = 0.44$; 95% CI [0.34, 0.54]; $p < 0.001$; Figure 2A [↗](#)). A similar result was obtained when outliers were removed in a sensitivity analysis (Hedges' $g = 0.38$; 95% CI [0.29, 0.46]; $p < 0.001$). Both the main analysis and sensitivity analysis showed lower bounds well above conventional thresholds for small effects ($g \geq 0.2$), thus indicating robust evidence (30) for meaningful improvement in episodic memory performance due to HITS.

Effects of HITS were greater on episodic-memory than non-memory tasks

Meta-regression was used to separately evaluate the overall effect of HITS on performance for 113 effects that concerned tasks assessing cognitive functions other than episodic memory. These included tasks designed to assess attention, working memory, executive functions, and language. The overall effect was almost zero and not significant, indicating that HITS did not reliably affect performance in these tasks (Hedges' $g = 0.04$, 95% CI [-0.01, 0.09], $p = 0.12$; Figure 2B [↗](#)). A similar result was obtained when outliers were removed in a sensitivity analysis ($g = 0.01$, 95% CI [-0.05, 0.06]; $p = 0.79$).

To more rigorously assess whether effects of HITS for tasks that measured episodic memory were significantly greater than for non-memory tasks, we ran a model pooling all 253 memory and non-memory effects and assessing overall moderation by task type. Effects for non-memory tasks were significantly less than for episodic-memory tasks (Hedges' g modification = -0.39; 95% CI [-0.48, -0.29]; $p < 0.001$). A similar result was obtained when outliers were removed in a sensitivity analysis (Hedges' g modification = -0.34; 95% CI [-0.43 - 0.25]; $p < 0.001$). Thus, the beneficial effects of HITS were significantly greater for tasks measuring episodic memory than other cognitive domains in both the main and sensitivity analyses, with no evidence that HITS affected tasks measuring other cognitive domains.

Study factors modulated HITS effects on episodic memory

We categorized episodic-memory effects according to study design factors, based on *a priori* hypotheses as well as *post hoc* considerations of variability in study designs (Table S1 [↗](#)). We evaluated which of these factors had the potential for independent modulation of HITS effects on episodic memory via cross-validated lasso regression selection with permutation. Factors that were retained in greater than 50% of the lasso iterations were included in the meta-regression. This led to the exclusion of the TMS Protocol factor, likely because there was strong redundancy of this factor with the Intensity and Sessions factors (Figure S2 [↗](#)). As expected, the 10 retained factors were well represented among analyzed effects (Figure 3 [↗](#)). In the sensitivity analysis with outlier removal, TMS Protocol was also discarded, as well as the Task Delay factor. Three of the retained factors significantly modulated the effects of HITS on episodic memory performance (Figure 3A [↗](#)), as described next. Consistent with our *a priori* hypothesis, effects for tasks that measured episodic memory using recollection test formats were significantly greater than for those that used recognition formats (Hedges' g modification = 0.23 greater for recollection versus recognition; 95% CI [0.08, 0.39]; $p = 0.004$; Figure 3A [↗](#)). In contrast, effects for tests that used other episodic memory (non-recollection and non-recognition) formats were not significantly different from recognition (Hedges' g modification = 0.07 less for other-format versus recognition; 95% CI [-0.33, 0.20]; $p = 0.62$; Figure 3A [↗](#)). The same pattern held in the sensitivity analysis with outliers removed (Hedges' g modification = 0.20 greater for recollection than recognition; 95% CI [0.07,

Study #	First Author	Year	Sample Size	DOI
<i>Included in meta-analyses</i>				
1	Wang	2014	16	10.1126/science.1252900
2	Yazar	2014	69	10.1371/journal.pone.0110414
3	Wang	2015	*	10.1002/hipo.22416
4	Bonni	2015	30	10.1016/j.bbr.2014.12.032
5	Nilakantan	2017	16	10.1016/j.cub.2016.12.042
6	Yazar	2017	23	10.1016/j.brs.2017.02.011
7	Kim	2018	32	10.1126/sciadv.aar2768
8	Tambini	2018	22	10.1162/jocn_a_01300
9	Bonnici	2018	22	10.1523/JNEUROSCI.1239-18.2018
10	Koch	2018	14	10.1016/j.neuroimage.2017.12.048
11	Wynn	2018	19	10.1101/lm.048033.118
12	Ye	2018	18	10.1523/JNEUROSCI.0660-18.2018
13	Hermiller	2019	24	10.1002/hipo.23054
14	Hermiller	2019	14	10.1002/brb3.1393
15	Nilakantan	2019	15	10.1212/WNL.00000000000007502
16	Hermiller	2020	16	10.1523/JNEUROSCI.0486-20.2020
17	Chen	2020	13	10.18632/aging.202313
18	Gao	2021	32	10.1016/j.brainres.2021.147510
19	Hebscher	2021	20	10.1016/j.cub.2021.01.027
20	Freedberg	2021	23	10.1016/j.neuroimage.2021.118199
21	Velioğlu	2021	15	10.1016/j.nlm.2021.107410
22	Jia	2021	69	10.3389/fnagi.2021.693611
23	Freedberg	2022	29	10.1016/j.bbr.2021.113707
24	Dave	2022	16	10.1016/j.crneur.2022.100030
25	Hua	2022	39	10.3389/fnhum.2022.973298
26	Hermiller	2022	30	10.1016/j.neurobiolaging.2021.09.018
27	Chen	2022	8	10.1073/pnas.2113778119
28	Wei	2022	29	10.1016/j.psychres.2022.114721
29	Tang	2023	89	10.1093/schbul/sbad015
30	Chen	2023	18	10.1111/cns.14177
31	You	2023	34	10.2147/CIA.S416992
32	Jin	2024	11	10.1016/j.bspc.2023.105725
33	Cheng	2025	15	10.1162/jocn_a_02273
34	Jung	2024	30	10.1001/jamanetworkopen.2024.9220
35	Lv	2024	60	10.1016/j.bbr.2024.115117
36	Webler	2024	24	10.1016/j.bpsgos.2024.100309
37	Velioğlu	2024	12	10.29399/npa.28420
38	Zheng	2024	43	10.1093/cercor/bhae460
<i>Excluded from meta-analyses</i>				
39	Zhao	2016	30	10.18632/oncotarget.13060
40	Warren	2019	*	10.7554/eLife.49458
41	Hendrikse	2020	39	10.1016/j.cortex.2020.08.028
42	Wang	2020	8	10.3389/fnhum.2020.541791
43	Cash	2022	*	10.1016/j.brs.2022.09.004
44	Yang	2022	16	10.3233/JAD-215390
45	Chen	2022	12	10.3233/JAD-210661
46	Li	2023	28	10.3390/brainsci13030419
47	Li	2024	61	10.2147/NDT.S468219
48	Koen	2018	20	10.1080/17588928.2018.1484723

Table 1. Studies identified by systematic review.

Study information is listed separately for studies that were included versus excluded from meta-analyses. Sample sizes listed are for the number of subjects contributing to statistical effects that were analyzed or described. *Studies that re-analyzed data from another study and therefore did not contribute independent data are indicated.

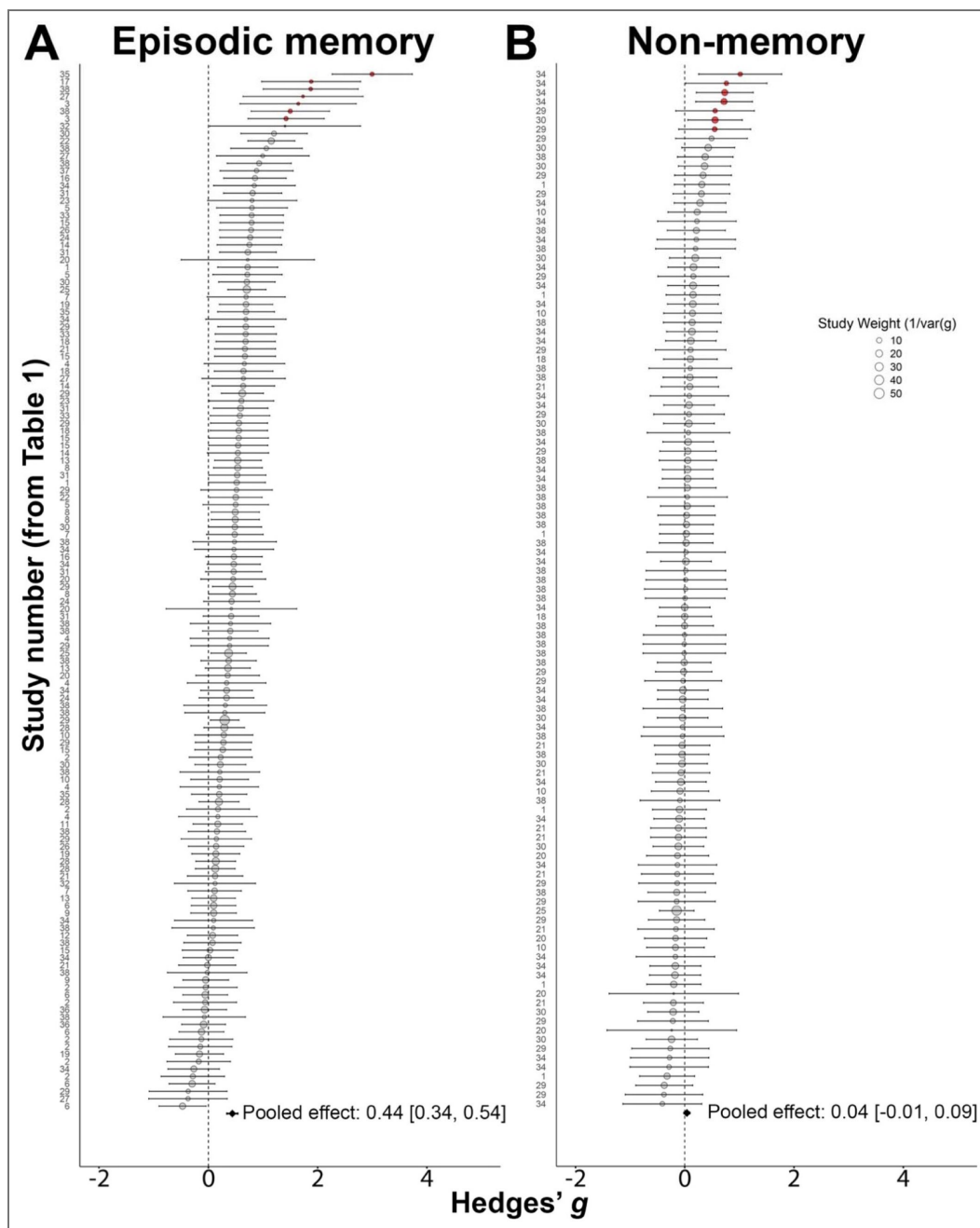


Figure 2. HITS selectively improved episodic memory overall.

(A) Forest plot of all effects of HITS on episodic-memory task outcomes, ordered by size. (B) Forest plot of all effects of HITS on non-memory task outcomes. The pooled effect is shown for both plots. Red circles indicate outliers that were excluded in sensitivity analyses. Circle size indicates the study's weighted contribution to the meta-regression model.

0.34]; $p = 0.003$; Hedges' $g = 0.01$ less for other-format than recognition; 95% CI [-0.24, 0.23]; $p = 0.96$). Thus, among tasks assessing episodic memory, the effects of HITS were significantly greater for recollection.

Regarding the Timing factor, HITS applied before the task (from days to seconds before) was associated with significantly greater effects on episodic memory than HITS applied immediately after the period of memory encoding and before the retrieval test (Hedges' g modification = 0.68 greater for pre-task versus post-encoding; 95% CI [0.34, 1.02]; $p < 0.001$; [Figure 3A](#)). The same pattern held in a sensitivity analysis with outliers removed (Hedges' g modification = 0.59 greater for pre-task versus post-encoding; 95% CI [0.32, 0.87]; $p < 0.001$). The average effect for studies in which HITS was applied post-encoding was nearly zero ([Figure 3A](#)), with these being the majority (52.4%) of all observed negative effects (i.e., effects reflecting impairment by HITS). In contrast, effects when HITS was applied pre-task were almost all positive (91.5% of pre-encoding effects were positive). Notably, studies in the post-encoding category all followed the same general experiment design, whereby a brief encoding period was immediately followed by a single session of stimulation which was then immediately followed by the test. In contrast, pre-task studies included a variety of designs, including those in which brief stimulation trains were given immediately before individual memoranda as well as those in which many days or weeks of stimulation were given for days to weeks before memory encoding ([Table S1](#)). Notably, this variability in the number of sessions was explicitly assessed via the Sessions factor ([Table S1](#)). Thus, among tasks assessing episodic memory, the effects of HITS were significantly greater when it was applied sometime before the task, and thus before the period of memory encoding.

Regarding the Targeting factor, effects differed for studies that used individualized (i.e., subject-specific) measures of functional or structural MRI connectivity to determine the stimulation location for HITS compared to studies that used a similar target for all subjects based on an atlas or *a priori* hypothesis (Hedges' g modification = -0.30; 95% CI [-0.54, -0.06]; $p = 0.013$; [Figure 3A](#)). However, this effect was not significant in a sensitivity analysis with outliers removed (Hedges' g modification = -0.16; 95% CI [-0.36, 0.03]; $p = 0.09$). Thus, evidence for modification of HITS effects by targeting method was weak, not surviving sensitivity analysis.

None of the other study factors significantly modulated the effects of HITS in either the main or sensitivity analyses ([Figure 3B](#)).

Of the 140 effects concerning episodic memory, 17.9% involved both the two significant positive moderating study factors identified in the main and sensitivity analyses (recollection test format and application of HITS before encoding). These effects thus can be considered as reflecting optimized experiment designs. To estimate the expected effects for future studies incorporating both of these factors, we used our meta-regression model to project the expected effect sizes for studies incorporating these factors while accounting for variance associated with other factors, held constant. Their mean effect estimate was $g = 0.66$ (95% CI [0.51, 0.82]), in contrast to that of all other effects incorporating non-optimized factors of all other effects, $g = 0.36$ (95% CI [0.29, 0.44]) ([Figure 4](#)).

Similar outcomes in studies that were excluded from meta-analysis

Ten studies identified via systematic review were not included in the meta-analysis for reasons summarized in Supplementary Results. Excluded studies reported outcomes from 214 subjects and incorporated study design factors representative of those for included studies (Supplementary Results). Eight of the excluded studies reported results from statistical tests to assess the effects of HITS on episodic memory performance. Of those, five reported statistically significant improvement due to HITS in at least one test of episodic memory, two reported numeric improvements due to HITS that were not statistically significant, and one reported equivalent performance for HITS versus a control condition (Supplementary Results). Thus, excluded studies mostly reported episodic memory improvements due to HITS, with variability in outcomes that is

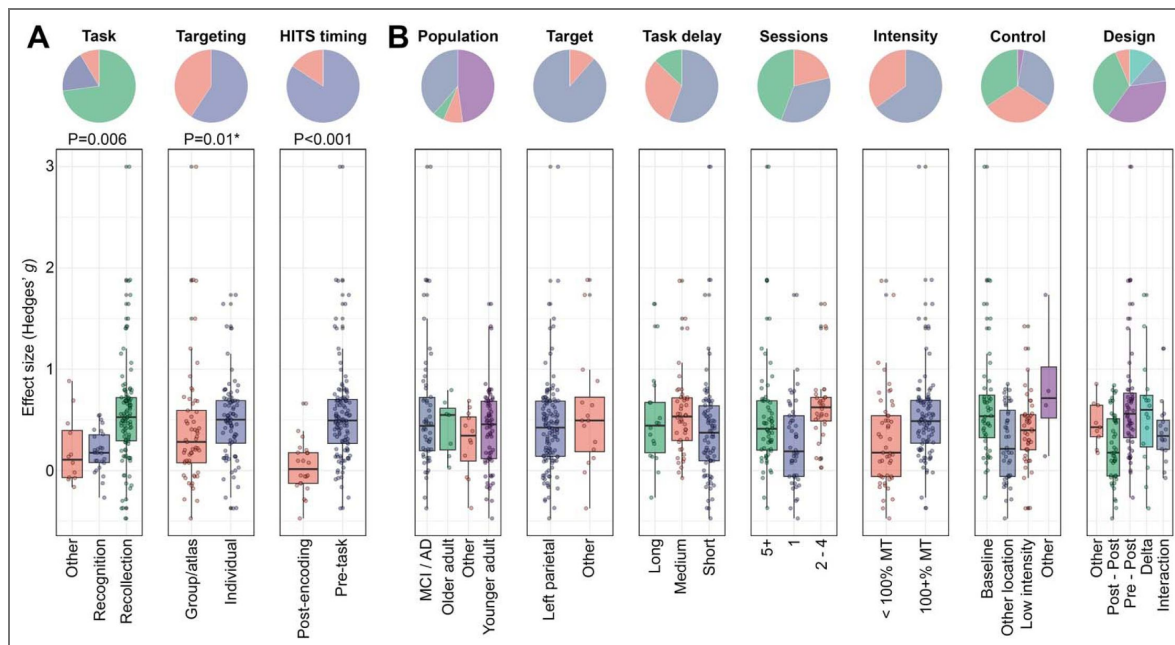


Figure 3. Study factors that modulated effects of HITS on episodic memory.

(A) Factors that significantly modulated HITS effects on episodic memory. The pie chart indicates the percentage of effects for each category of each factor. Individual effects are shown in the box plots, grouped by factor levels. (B) Factors that did not significantly modulate HITS effects on episodic memory, plotted in the same format. *Indicates an effect modification that was significant in the main analysis but not in the sensitivity analysis.

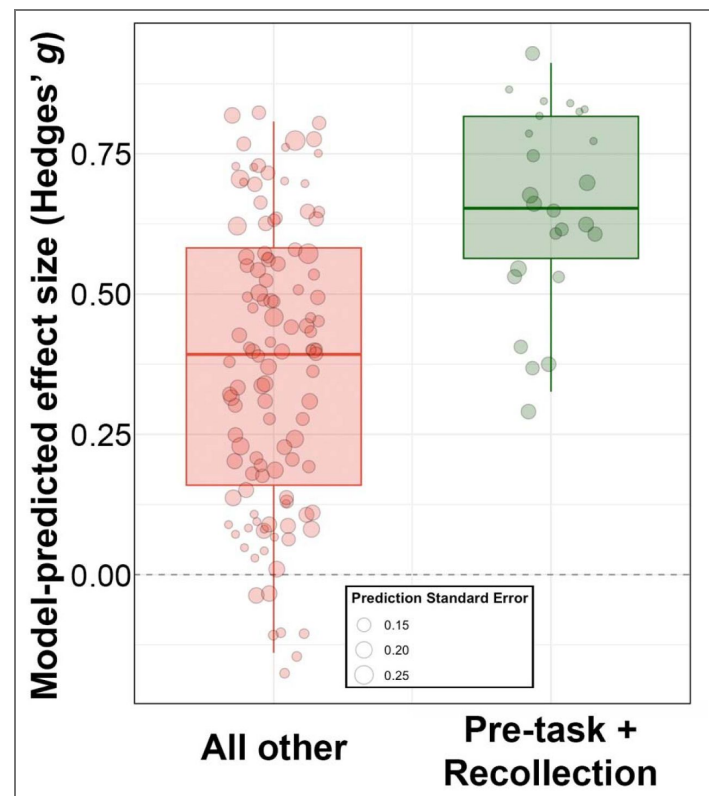


Figure 4. Greater effects of HITS on memory in studies having optimized designs.

Box plots of expected effect sizes projected from the meta-regression model for studies in which HITS was applied pre-task and measured using recollection format tests, versus all other study factors. Circles are individual projected effects, with circle size indicating standard error of the prediction, as indicated.

consistent with the variability seen in studies that were included in the meta-analysis. It is therefore unlikely that meta-analysis results would vary significantly if it were possible to include these studies.

No serious adverse events were reported

None of the reviewed studies reported any unexpected or serious adverse events. The total sample reviewed included N=1,223 subjects, including healthy younger and older adults, individuals with mild to moderate Alzheimer's dementia, and individuals with psychiatric symptoms. This suggests that the incidence of serious adverse events due to HITS is low, which is consistent with the general evidence for very low incidence of serious adverse events due to rTMS (31).

Discussion

Enhancement of episodic memory in humans is an important scientific objective with clear implications for the treatment of clinical memory disorders and for enhancing function in healthy individuals, including older adults experiencing normative memory decline. However, interventions for memory enhancement have been elusive. Our findings indicate HITS is a promising method to achieve memory enhancement via noninvasive stimulation. There were similarly positive effects among participant populations, from young healthy adults to those with memory disorders, and for a wide range of assessment delays, from seconds to weeks after subjects received HITS. In contrast, there was essentially no effect on non-memory functions, suggesting that memory enhancement did not come at the cost of any off-target cognitive detriment. There were no reported unexpected or serious adverse events. Findings suggest that HITS can be used to achieve robust and selective enhancement of episodic memory.

Many have speculated that brain stimulation targeting a given network can have specific effects on the cognitive abilities hypothesized to depend on that network (17, 22, 27, 32, 33). Although aspects of this hypothesis for HITS have been supported in isolated experiments (8, 34, 35), the current findings are notable in providing meta-analytic support. The hippocampus and its network are strongly implicated in episodic memory, and particularly in tests that measure recollection compared to recognition (36–38). Indeed, we found that effects of HITS were specific to episodic-memory versus other cognitive tests and were greater for recollection than recognition or other-format memory tests. Although previous individual experiments have shown that HITS affects activity of the hippocampal network in relation to improved recollection more so than recognition (25, 35, 39), the current findings provide strong evidence that across many different experiment parameters, stimulation targeting the hippocampal network specifically affects the memory function thought to depend most heavily on this network.

Specificity of HITS effects on recollection is mechanistically informative. For instance, other interventions such as transcranial electrical stimulation yield nonspecific effects across a variety of cognitive domains, including when applied to the lateral parietal areas overlapping with those reviewed here (40). It is possible that such nonspecific effects could arise via modulation of general non-targeted factors, such as arousal via effects of stimulation on either peripheral or central nervous systems (41, 42). Such nonspecific influences could be problematic, as general and/or off-target effects could be ineffective or detrimental when applied to disorders affecting specific brain circuits. Further, TMS of frontal cortex developed for anti-depressant intervention has been reported to affect symptoms of a variety of psychiatric disorders and to nonspecifically affect performance of tasks measuring a variety of cognitive abilities (43). Along with previous neuroimaging evidence for selectivity (8, 25, 34, 35, 44, 45), the current findings suggest that HITS is suitable for targeted interventions of hippocampal network contributions to recollection. This could be useful in the treatment of memory disorders, which tend to disproportionately impact recollection relative to recognition (37). However, an important caveat is that the tests used as outcomes in the studies we analyzed were primarily designed to measure different cognitive abilities in isolation in the context of neuropsychological diagnosis. Our findings thus do not preclude off-target effects in specialized tasks that deliberately assess the interactions of brain networks. That is, enhancement of episodic memory could affect tests of other cognitive domains

when those tests are at least partially sensitive to, or compete with, episodic memory (46). Nonetheless, the current results support HITS as notable among existing neuromodulatory approaches for producing selective effects on the intended cognitive function, with robust evidence for reproduction of effects supported via meta-analysis.

It is notable that HITS had little to no effect in experiments that delivered a single session of HITS immediately after an encoding session and before the corresponding memory test (Figure 3A, HITS Timing factor of Post-Encoding). In contrast, effects were robustly positive when HITS was applied at some point before the task, and thus before memory encoding, across a variety of potential stimulation-encoding delay intervals. This suggests that HITS improves memory formation rather than retention of recently learned information or memory retrieval. This is consistent with findings that hippocampus makes distinct contributions to learning versus retrieval (47). However, an important caveat to this interpretation is that studies did not attempt to cleanly dissociate effects of HITS on encoding versus retrieval, and so any interpretation of selectivity of HITS effects on encoding should be made with caution. Regarding HITS as a potential intervention, it is notable that effects were similar across a variety of delays between when HITS was applied and when memory encoding occurred (Figure 3B, Task Delay factor). Unlike potential interventions that require delivery of stimulation concurrent with cognitive performance (e.g., transcranial electrical stimulation) or with highly specific information about the timing or successfulness of specific memory events (e.g., closed-loop brain stimulation), this finding suggests that HITS does not require specific knowledge of when memory demands occur following intervention. This is of practical utility, as it is difficult to know *a priori* when memory encoding will be required during ongoing activities of daily living.

It is important to avoid over-interpretation of null findings concerning moderating effects of some study factors (Figure 3B), as many were correlated (Figure S2) and were selected in individual experiments based on *a priori* hypotheses. For instance, based on evidence that stimulation for more consecutive sessions of HITS leads to greater or more persistent effects (48), studies that included more HITS sessions also tended to measure outcomes after longer delays (correlation of Sessions and Task Delay factors; Figure S2). Thus, it is possible that studies with more stimulation sessions would have yielded greater effects had they used shorter testing delays, as in the studies with fewer stimulation sessions. Likewise, the Intensity factor was complicated in that studies using lower intensity were more likely to involve theta-burst stimulation than studies using higher intensity, which typically used high-frequency rTMS. These studies were based on hypotheses that hippocampal networks should respond to theta-burst stimulation more than high-frequency stimulation (22, 45). However, theta-burst stimulation typically used lower stimulation intensity and also fewer pulses of stimulation than high-frequency rTMS. Thus, no differences in outcome based on the Intensity factor could support the hypothesis that theta-burst stimulation is more effective than rTMS, as similar effect sizes resulted with fewer and less-intense stimulation pulses. Interpretation of null effect modulation by factors that are highly correlated with other factors is thus ambiguous. Although it was useful to identify those factors that robustly modulated effects of HITS via meta-analysis (Figure 4), future experimental studies will be needed to systematically assess whether effects of HITS vary by study design (Figure 3B), based on specific manipulation of individual factors while holding other factors constant.

A limitation of our approach is that we focused on studies applying HITS to lateral and medial parieto-occipital neocortex targets of the hippocampal network. Other prominent locations of this network include portions of medial frontal and lateral temporal neocortex (49). However, no studies targeting these locations were identified by our review, potentially because electrical fields generated by conventional TMS are not sufficiently intense or focal or because stimulation is not well tolerated for these targets. Notably, the hippocampal network includes portions of lateral cerebellar cortex (50), and one study identified by our review found episodic memory improvement following theta-burst TMS of this location (51). Theta-burst stimulation of lateral cerebellum has also been shown to affect hippocampal network fMRI connectivity (52). However,

this study was excluded from meta-analysis to avoid heterogeneity in stimulation locations. Future work could systematically evaluate alternative stimulation targets within the hippocampal network, including those that may require stimulation methods other than TMS.

The current findings suggest that HITS has robust effects on episodic memory and that studies including optimized parameters yield especially robust effects (Figure 4). However, further research would be needed to test clinical efficacy for any specific conditions and to determine mechanisms of action. Although our review included studies of patients with episodic memory impairments due to mild to moderate dementia and psychiatric symptoms, the studies did not assess whether gains by HITS could be maintained for months to years or whether they were clinically meaningful. It is also not clear whether HITS will be useful for memory impairments in disorders other than those included in our review, which may involve distinct mechanisms, or whether there is a threshold for degradation of functional and/or structural brain networks beyond which HITS would no longer be effective. We consider our findings as motivating for future attempts to evaluate HITS safety and efficacy for specific clinical indications. Implementations for specific disorders will likely require better understanding of how HITS affects neural function, such as could be obtained by directly recording hippocampal activity while HITS is applied (45, 53), in order to tailor HITS to target disorder-specific neural impairments. The current findings encourage such future explorations.

Materials and methods

Eligibility criteria

We included reports published between September 2014 (date of the seminal HITS publication (8)) and April 2025 that: (1) used rTMS to stimulate hippocampal network locations and (2) included at least one objectively scored test of episodic memory as an outcome. Studies were considered as stimulating hippocampal network locations if they delivered rTMS to: (1) an individualized/subject-specific location defined by fMRI connectivity analysis (e.g., a neocortical location identified by resting-state fMRI connectivity to hippocampus in an individual), (2) a location based on a group metric of hippocampal fMRI connectivity (e.g., a neocortical location identified in a group-based atlas of hippocampal resting-state fMRI connectivity), and (3) a similar location based on a hypothesis (e.g., a location that is in a group atlas of hippocampal connectivity but that was selected to test a hypothesis about its function). We limited spatial variability among studies by including only those that stimulated parietal and superior occipital neocortex. We excluded studies using other potential hippocampal network locations due to their rarity, including one study with cerebellar cortex stimulation (51). Studies were considered as including an objective test of episodic memory if at least one of the reported outcomes used a memory test that allowed unambiguous computation of the effect of stimulation on performance accuracy (i.e., experimenter-controlled stimulus set presented to subjects for encoding, for which memory was subsequently tested after some retention interval). When studies also included tests of episodic memory that could not be unambiguously scored (i.e., subjective reports that could not be verified for accuracy), only the objectively scorable tests were included in our analysis.

Search strategy

Two publicly available online scientific databases were searched: PubMed (pubmed.ncbi.nlm.nih.gov) and Scopus (scopus.com). These databases were most recently searched on April 26, 2025. The primary search was conducted with PubMed, using the PubMed Advanced Search Builder. Two search terms of interest were entered: “memory” and “transcranial magnetic stimulation”. These terms were selected to be as broad as possible while aiming to exclude studies utilizing stimulation techniques (invasive and non-invasive) other than TMS. These terms were added to the Query Box using ‘AND’ Boolean operators; all possible fields (i.e., title, abstract) were searched. PubMed automatically included a search for similar terms for each of the terms of interest entered (i.e., ‘memory’ and ‘memories’) and included the Medical Subject Headings (MeSH) terms associated with them. This search returned 104 results. A secondary

search was conducted using Scopus after completing the PubMed search, using the same primary search terms using ‘AND’ Boolean operators and constraining fields to Article title, Abstract, and Keywords and limiting results to peer-reviewed empirical reports by limiting the “Document Type” option to “Article”. This search returned 160 results. We also examined the references of publications identified as meeting the eligibility criteria for suggestions about other relevant publications. Publications were selected from the search results when two reviewers (PFA and JLV) agreed that they met the two eligibility criteria. Additionally, only studies in human participants were selected.

Outcomes extraction

Following review of the full texts, we extracted data from statistical tests aiming to quantify the effects of rTMS on episodic memory and non-memory outcomes. Extracted data included sample sizes, means, and standard deviations of the outcome variables for experimental and control measurements, as well as results from reported statistical tests. These data were used to calculate effect sizes, using the test statistics or group means and variances following established methods (54). Cohen’s *d* was used to measure the effect size of pairwise comparisons. Partial eta squared was used to measure the effect size of comparisons using analysis of variance or other general linear models. We did not include effects from tests that provided “composite” measures spanning multiple cognitive domains (e.g., Montreal Cognitive Assessment, Mini-Mental State Examination, etc.) as these could not be readily classified as reflecting episodic-memory versus various non-memory functions. In tasks with multiple subscales (e.g., Wechsler Memory Scale, NIH Toolbox for Cognition, etc.), individual subscale scores were used and categorized based on the task format (e.g., recall versus recognition) and/or by the intended cognitive domain (e.g., episodic memory versus sustained attention). Effect sizes were computed such that improvement in performance (i.e., higher accuracy, faster response time in speeded non-memory tasks, or fewer errors) were positive and impairments (i.e., lower accuracy, slower responses, or more errors) were negative. Prior to input for meta-regression, all estimates of effect size were converted to Hedges’ *g* to minimize potential impact of inflated effect sizes associated with small sample sizes (54). Within and between estimation uncertainty around each Hedge’s *g* were approximated using its corresponding study effective sample size.

Meta-analysis of effect size

A mixed effects univariate meta-regression model estimated via restricted maximum likelihood was used to estimate the overall effect of rTMS on outcomes, pooling all effects reported in the final sample of studies collected at the outcome level and weighting them in terms of their within and between estimation uncertainty. We included a study random effect in the model to account both for inter and intra study variability in our final estimate, as a single study often reported various effect sizes for different outcomes. Heterogeneity in individual effect sizes was examined using a generalized Q-test, to assess whether variability in observed effects exceeds what would be expected from random sampling error alone (55, 56).

Meta-analysis of effect modifiers

Factors that varied among studies were identified based on *a priori* hypotheses and *post hoc* consideration of study design details that could influence outcomes (Table S1 [↗](#), Figure 3 [↗](#)). Our categorization of factors was intended to include important design issues that could inform future interventional trials, such as the study population investigated, whether experiments used multi-session stimulation with interventional goals versus acute stimulation with basic-science goals, the delay at which outcomes were assessed, and the trial design and control conditions. The accompanying supplementary table describes these factors and the rationale for their inclusion (Table S1 [↗](#)).

We explored the association of effect size with these suspected modifiers by estimating a multivariate meta-regression model including all study factors. As with the main effects model, we pooled all effects reported in the final sample of studies collected at the outcome level and

weighted them in terms of their within and between study estimation uncertainty. Since some study characteristics were *a priori* expected to be highly correlated with one another (e.g., theta-burst stimulation is typically performed at lower intensity than high-frequency rTMS), we performed a previous-variable-selection step before assessing effect modification. Indeed, some factors were highly correlated, as expected (Figure S2 [↗](#)). We therefore used a bootstrapped (1000 iterations) lasso model with a cross-validated penalization term to identify study characteristics that were jointly consistently redundant in terms of statistically explaining HITS effects (i.e., discarded by lasso with a probability larger than 50% across all iterations), to be then excluded from the effect modification analysis. Study factors retained after the lasso variable selection were then simultaneously included in the multivariate mixed effects meta-regression, where their association with stimulation effects was estimated. Heterogeneity in individual effect sizes after accounting for effect modifiers was examined using a generalized Q-test, assessing whether significant variability in observed effects remained, even after accounting for the variation explained by included study characteristics ($Q(df = 136) = 307.07, p < 0.001$). Analysis for this step was conducted using the *glmnet* package in R. All meta-analytic estimates were computed using the *metafor* package in R ([56](#)).

Outlier removal

To guard against the impact that unusually strong reported effects could have had in our findings, we repeated all meta-analyses, including the lasso variable selection step, after removing outliers identified using the studentized deleted residuals ([57](#)). We identified and excluded seven effect outliers in the analysis of episodic memory outcomes, leaving 133 of 140 effects for that analysis, and seven effect outliers in the analysis of non-memory outcomes, leaving 106 of 113 effects for that analysis. Categorical variables were included in final models if at least one level had a coefficient with magnitude $\geq |0.05|$ or their aggregate coefficients had magnitude $\geq |0.01|$. Variables with all coefficients $< |0.01|$ were not included. We report findings with and without removal of outliers, which had little impact on overall results.

Reporting bias assessment

Publication bias was assessed with a funnel plot of individual effect sizes against their standard errors (Figure S3 [↗](#) and S4 [↗](#)). Special attention was paid to the asymmetry of the plot and the absence of effects in the lower-left corner of the funnel, which indicates the lack of published small-sample studies reporting negative effect sizes, further confirmed by an Egger's regression test ($p < 0.001$), which is commonly carried out to detect evidence of publication bias ([58](#)). We tested for heterogeneity (I^2) to measure the inconsistency of effect sizes across studies within each cognitive domain. The I^2 statistic represents the percentage of variation in effect sizes beyond a random sampling error ([59](#)). Although the interpretation of heterogeneity can depend on multiple factors, an I^2 of 25 to 75% is typically considered moderate, whereas $I^2 > 75\%$ is considered a substantial heterogeneity ([59](#)). Further, we tested for the presence of outliers (described above), and removed them in sensitivity analyses to assess the robustness of our main findings to outliers. There was no conclusive evidence of publication bias, with I^2 of 0.53 for episodic memory tests and of 0.0 for non-memory tests (Figure S3 [↗](#) and S4 [↗](#)) and with replication of major findings in the sensitivity analyses.

Supplementary Materials

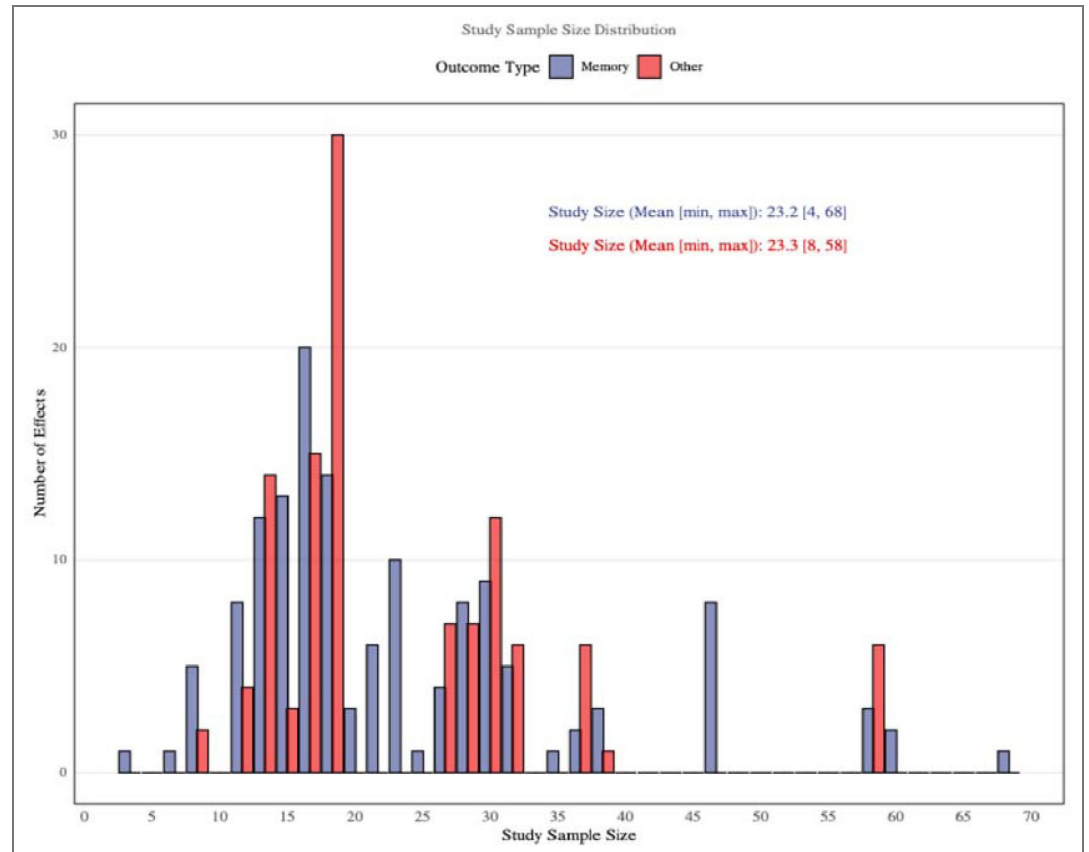


Figure S1. Histogram of sample sizes per analyzed effect, plotted separately for episodic-memory and non-memory effects.



Figure S2. Relationships among factors in episodic memory studies.

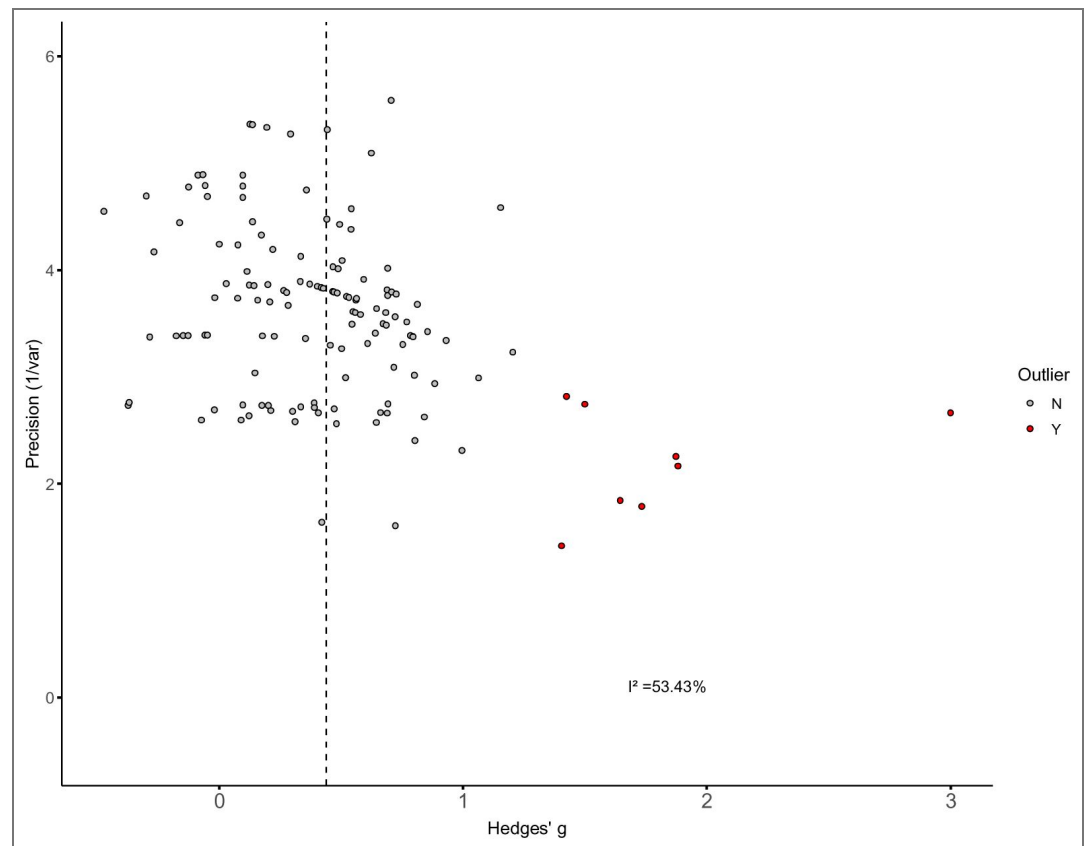


Figure S3. Funnel plot of episodic memory HITS effects. Outliers are indicated in red. The I^2 statistic is indicated.

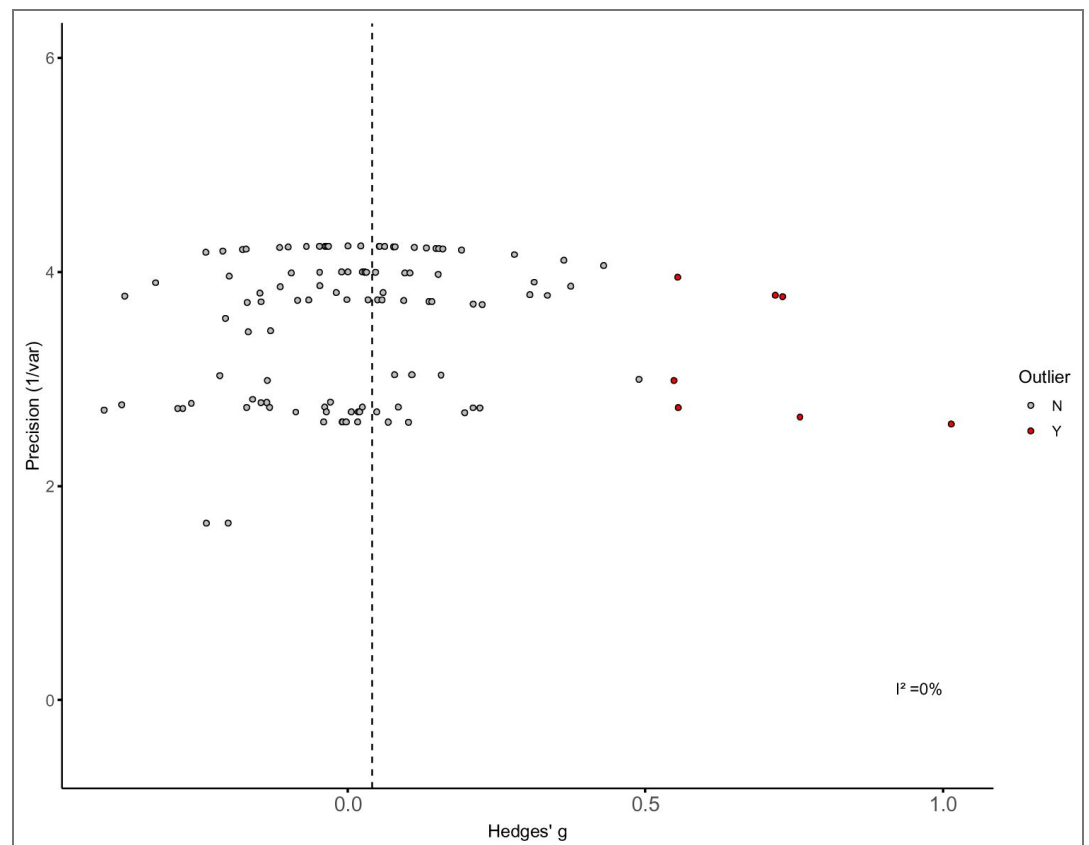


Figure S4. Funnel plot of non-memory HITS effects. Outliers are indicated in red. The I^2 statistic is indicated.

#	Factor	Levels	Description	Rationale
1	Task	Recognition	Episodic memory tests using recognition formats; i.e., memoranda are repeated from study to test and subjects make old/new discrimination judgments	Recollection is thought by many to be more heavily dependent on hippocampus and its network than familiarity, which differentially contribute to tasks using recollection and recognition formats. While Recollection tasks could have been sensitive to many types of memory processing, including familiarity, at least one aspect of performance depended on recollection. Performance of tasks with other formats could be supported by an unknown mix of recollection, familiarity, or other processes.
		Recollection	Episodic memory tests using recollection formats; i.e., free recall, cued recall, or self-reported recollection judgments for tasks using paired associates, item-context associates, or multi-part complex stimuli as memoranda	
		Other	Episodic memory tests using other formats; i.e., temporal order judgments, recall of details from videos, scored visual reproduction, and tests that assess mnemonic pattern separation	
2	Targeting	Individual	The location where TMS was applied was determined by subject-specific analysis of functional or structural MRI connectivity of hippocampus; i.e., a location of parietal/occipital neocortex was targeted based on a maximal connectivity value with hippocampus defined in each subject.	Individualized targeting has been used in many studies based on the assumption that network structure varies across people and that indirect influences of stimulation on the hippocampus are transsynaptic and predicted by connectivity. Group targeting assumes that there is sufficient homogeneity across subjects to reliably affect networks, or was based on hypotheses about specific parietal areas that happen to be among hippocampal network areas typically targeted using individualized targeting approaches.
		Group/atlas	TMS was applied to approximately the same target in each subject, based on functional or structural MRI connectivity in a group/atlas or based on a specific hypothesis about the function of a specific location.	
3	HITS Timing	Pre-task	TMS was delivered sometime before the period of memory encoding, ranging from application during the several seconds before individual memoranda were studied, in the minutes to hours before an encoding block was administered, to for many sessions over several days or weeks before the encoding phase of the memory task was completed.	This is a post hoc factor. Most studies used the "before" design whereas some used the "after" design. Hippocampal involvement in memory could differ for encoding versus retrieval, although an important caveat is that the timecourse of stimulation effects on brain activity and memory is not known, and so claims of temporal specificity cannot be assessed.
		Post-encoding	All of these experiments applied a single session of stimulation after a block of encoding and before the corresponding memory test was given minutes to tens of minutes later.	
4	Population	Younger adult	Young adults with no reported neurologic or psychiatric disorders were participants (<45 years old).	Populations with age-related memory decline (older adult) or with clinical memory impairments (MCI/AD and potentially Other) could present more opportunity for stimulation to improve memory. Alternatively, network disruptions frequently observed in memory disorders could reduce the ability for TMS to indirectly influence hippocampus due to reduced possibility of transsynaptic influence. Thus, the a priori hypotheses are that effects will differ among populations.
		Older adult	Older adults with no reported neurologic or psychiatric disorders were participants (>65 years old).	
		MCI / AD	Participants included individuals with diagnoses of Mild Cognitive Impairment (MCI) or mild to moderate Alzheimer's dementia (AD) following clinical standards.	
		Other	Other groups included individuals with first-episode psychosis, individuals at high risk of schizophrenia, and individuals with symptoms of anxiety disorders of post-traumatic stress disorder.	
5	Target	Left parietal	Lateral parietal cortex (LPC) was stimulated in left hemisphere in most studies, at locations defined via either individualized or group targeting	This is a post hoc factor. The majority of studies targeted areas of lateral, inferior, posterior parietal cortex near the angular gyrus, following the

		Other	methods, as shown in Figure 1. Other stimulation locations included precuneus or other parietal-occipital areas defined based on individual or group targeting methods, as shown in Figure 1. These were combined because separating them would have led to very few effects in each separate level.	general method of Wang et al. 2014. Others targeted precuneus or other parietal or anterior occipital areas based on connectivity with hippocampus. All of these areas are considered within the hippocampal network and so no strong a priori hypotheses regarding the effects of HITS were warranted.
6	Task delay	Short	The effects of stimulation were measured on the same day on which stimulation was delivered. Some studies applied short stimulation trains in the seconds before memoranda were presented, others delivered a several-minute stimulation train and then tested effects within one hour, and others used multi-day or multi-week stimulation and tested the effects on the same day following the last stimulation session.	This is a post hoc factor. Generally, the effects of stimulation are thought to decay, albeit at unknown rates. However, studies tended to measure outcomes at test delays aligned with the duration of stimulation (i.e., shorter test delays for protocols involving less stimulation and longer test delays for protocols involving more stimulation over more days; see Figure S2). Thus, as test delay was aligned to measure outcomes appropriately for hypothetical protocol-specific hypothesized decay rates, there was no strong a priori hypothesis to suggest that test delays would be non-equivalent.
		Medium	The effects of stimulation were measured on the following day after stimulation delivery. These studies mostly delivered several consecutive daily sessions of stimulation and measured the effects ~24 hours after the final stimulation session.	
		Long	The effects of stimulation were measured more than one day after the final stimulation session in multi-day or multi-week stimulation protocols, up to ~4 weeks after the end of stimulation.	
7	Sessions	1	Protocols involved only one session of stimulation before outcomes were assessed. This encompasses studies that used short trains of stimulation (seconds) immediately before memoranda were presented, as well as several-minute trains of stimulation before memory was assessed.	This is a post hoc factor. Generally, more sessions of stimulation are thought to produce greater and/or longer-lasting effects. However, as session counts and task delays were correlated (see "Task delay" factor), there were no strong a priori hypotheses about differences among session counts.
		2 – 4	Protocols involved two to four consecutive sessions of stimulation spread out across multiple days.	
		5+	Protocols involved five or more sessions of stimulation spread out across multiple days and/or weeks	
8	Intensity	100+% MT	Stimulation was applied at an intensity of greater than or equal to the resting motor threshold (MT), typically no higher than 120% of the MT.	This is a post hoc factor. Generally, higher intensities would be thought to have greater effects. However, this factor overlapped considerably with TMS protocol (factor 11, which was not retained in the lasso regressions). That is, studies using repetitive TMS trains at high frequencies tended to use intensities at or above motor threshold, whereas studies using patterned theta-burst stimulation tended to use intensities lower than 100% of motor threshold (Figure S2). Thus, there were no strong a priori hypotheses concerning this factor specifically.
		<100% MT	Stimulation was applied at an intensity of less than 100% of the resting MT, typically at 90%, 80%, or 70% of the MT. Note that one study (study 8) used 80% of active MT, rather than resting MT.	
9	Control	Baseline	The comparison for the effects of stimulation was the baseline, pre-stimulation performance value.	This was a post hoc factor. Different control conditions are used to achieve different scientific objectives. As all control conditions would be expected to yield little if any effect on memory performance, any significant variation in effects of HITS on memory by control condition would raise doubts
		Other location	The comparison for the effects of stimulation of the hippocampal network was stimulation of some other part of the brain not within the hippocampal network using the same stimulation parameters.	

			Typically, primary or secondary motor or somatosensory areas were used as control locations.	about its robustness to small differences in experiment design choices.
		Low intensity	The comparison for the effects of stimulation was stimulation of the same location using the same overall stimulation parameters but at a very low intensity, typically <30% of resting motor threshold.	
		Other	The comparison for the effects of stimulation was stimulation of the same location using a different rhythmic pattern that was hypothesized to have less of an effect on memory.	
10	Design	Post - Post	The effects of stimulation on performance were statistically compared for after receiving HITS versus after receiving the control condition (e.g., t test of the performance scores after HITS versus after control).	This was a post hoc factor. Different statistical designs are used to achieve different scientific objectives and to accommodate different practical constraints on experiment design. As the choice of statistical design would be expected to yield little if any effect on memory performance, any significant variation in effects of HITS on memory by analytic design would raise doubts about its robustness to small differences in experiment analysis choices.
		Pre - Post	The statistical comparison of effects considered performance after versus before receiving HITS (e.g., t test of performance scores after HITS versus baseline before HITS).	
		Delta	The statistical comparison of effects on performance was done by computing the pre- versus post-HITS change (delta) in scores and comparing to the pre- versus post-Control delta (e.g., t test of percentage change baseline to post-HITS versus percentage change baseline to post-Control).	
		Interaction	The statistical test of effects on performance treated time (pre- versus post-) and stimulation type (HITS versus control) as factors, as in a RM-ANOVA.	
		Other	More than two stimulation conditions, including the control condition, were compared as factors, as in a RM-ANOVA.	
11	TMS protocol	rTMS	rTMS (1, 10, 15, or 20 Hz) applied in either short trains of several seconds, minutes, or as a longer, approximately 20-min trains per session. Note that only two studies used 1 Hz rTMS whereas all others used 10Hz or higher.	Some evidence suggests that theta burst stimulation has greater effects on hippocampal network activity and episodic memory performance. However, this factor was strongly correlated with Intensity and Sessions factors (Figure S2). NOTE: this factor was not retained in the lasso regressions as it was strongly correlated with other factors, thus warranting no strong a priori hypotheses of its independent contributions.
		TBS	Theta-burst stimulation, including short trains of several seconds of theta burst, or as continuous theta burst (40 s) or intermittent theta burst (5 minutes) sessions.	

Table S1. Description of study factors.

Supplementary Results

Summary of findings from studies that were not included in the meta-analysis

As indicated in the main text, ten studies were excluded from meta-analysis (Table 1 [1](#)). Five of these exclusions were because statistical information provided in the publications were not sufficient for calculation of standardized effect sizes, and authors could not be reached to provide additional information (studies 39, 42, 44, 45, 47 from Table 1 [1](#)). One exclusion (study 40) was made because the sample of subjects analyzed overlapped 94% with subjects that contributed data to a previous publication that was included, and so exclusion was necessary to avoid redundant statistical results in the meta-analysis. Two studies were excluded because they did not report the effects of HITS on performance of an episodic memory test, but instead quantified the correlation between fMRI connectivity and the effects of HITS on episodic memory performance (studies 43 and 46). One study was excluded (study 41) because re-analysis of the same data by the same research team (study 43) reported a different conclusion than the original report, suggesting that stimulation may not have been delivered to intended hippocampal network targets in the original report, and thereby calling into question findings from the original report and making it unclear

how to categorize the Targeting or Target factors for meta-analysis. One study was excluded (study 48) because HITS was applied “online”, simultaneous with the trials in which memoranda were shown during the encoding period, whereas all other reviewed HITS studies used “offline” stimulation, with HITS applied sometime before or after the task period, and therefore it would have been problematic to categorize this study for the Timing factor.

Of the nine excluded studies, eight reported the effects of HITS on episodic memory performance (studies 39, 40, 41, 42, 44, 45, 47, 48). Five of these eight reported that HITS significantly improved performance in at least one of the episodic memory tasks that were administered (studies 39, 40, 42, 44, 45). One of those studies (study 44) also reported numeric but non-significant decrease in memory performance in a subgroup analysis of subjects with mild Alzheimer’s dementia. Two of the eight studies (41 and 47) reported small numeric improvements in memory following HITS that were not statistically significant. One of the eight studies (48) reported memory performance following HITS that was almost identical to that in the control-stimulation (vertex) condition. Notably, study 48 used “online” stimulation during memory encoding, which has generally been associated with performance impairment (e.g., References (61, 62)).

Regarding the study factors that modulated the effects of HITS in the main study (Figure 3A), the eight studies all included at least one test using a recollection format (Task factor: Recollection) and nine administered HITS prior to the memory task (Timing factor: pre-task). Two used individualized targeting (studies 40 and 41), with the others using group/atlas-based targeting. Other factors that were not significantly related to memory outcomes following HITS in the main analysis (Figure 3B) varied among these seven excluded studies.

The other two excluded studies (43 and 46) investigated correlations between resting-state fMRI connectivity and the effects of HITS on episodic memory, without directly assessing the impact of HITS on episodic memory performance. One reported that fMRI connectivity of the hippocampus to the stimulated location predicted episodic memory improvement due to HITS (study 43). The other study reported that HITS changed the correlation between fMRI connectivity and memory performance scores (study 46).

Overall, reported results in the excluded studies were generally consistent with findings from the meta-analysis of included studies. That is, of the eight excluded studies that reported effects of HITS on episodic memory, the majority (5/8) reported statistically significant episodic memory enhancement by HITS, but with variability in effect sizes among studies and a minority (2/8) reporting small but statistically nonsignificant improvement in memory due to HITS. Thus, if effects from these studies could have been included in the meta-analysis, it is unlikely that conclusions would have been substantially affected.

Data availability

A table of standardized effect sizes categorized by study and by the factor categories to which they were assigned for analysis is deposited in Dryad (datadryad.org), available for download at: <https://doi.org/10.5061/dryad.vhhmgqp86>.

Additional information

Author contributions

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Methodology: EBG

Investigation: PFA, JMHR, ASA, JLV

Writing – original draft: EBG, PFA, JLV

Writing – review & editing: EBG, PFA, JMHR, ASA, JLV

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Peer reviews

Reviewer #1 (Public review):

Summary:

Goicoechea et al. conducted a timely and thorough meta-analysis on the potential for indirect hippocampal targeted transcranial magnetic stimulation (TMS) to improve episodic memory. The authors included additional factors of interest in their meta-analysis which can be used to inform the next generation of studies using this intervention. Their analysis revealed critical factors for consideration: TMS should be applied pre-encoding, individualized spatial targeting improves efficacy, and improvement of recollection was stronger than recognition.

Strengths:

As mentioned previously, the meta-analysis is timely and summarizes an emerging set of studies (over the past decade since Wang et al., Science 2014). Those outside of the field may not be aware of the robustness in improvements in episodic memory from hippocampal targeted TMS. The authors were quite thorough in including additional factors which are important for the interpretation of these findings. These factors also address the differences in approach across studies. The evidence that individualized spatial targeting improves TMS

efficacy is consistent with recent advances in TMS for major depressive disorder. The specificity of the cognitive improvements to recollection of episodic memory and not for other cognitive domains is consistent with hippocampal targeting. The authors also plan to post the complete dataset on an open-source repository which enables additional analysis by other researchers.

Weaknesses:

The write-up is succinct and emphasizes the scientific decisions that underly key differences in the various experimental designs. While the manuscript is written for a scientific audience, the authors are likely aware that findings like this will be of broad appeal to the field of neurology where treatments for memory loss are desperately needed. For this reason, the authors could consider including a statement regarding an interpretation of this meta-analysis from a clinical standpoint. Statements such as 'safe and effective' imply a clinical indication and yet the manuscript does not engage with clinical trials terminology such as blinding, parallel arm versus crossover design, and trial phase. While the authors might prefer not to engage with this terminology, it can be confusing when studies delivering intervention-like five-days of consecutive TMS (e.g., Wang et al., 2014) are clustered with studies that delivered online rhythmic TMS which tests target engagement (e.g., Hermiller et al., 2020). While the 'sessions' variable somewhat addresses the basic-science versus intervention-like approach, adding an explicit statement regarding this in the discussion might help the reader to navigate the broad scope of approaches that are utilized in the meta-analysis.

Following revision: The authors have adequately addressed my concerns.

<https://doi.org/10.7554/eLife.108934.2.sa3>

Reviewer #2 (Public review):

Parietal lobe TMS, targeted to the episodic memory network via connections with the structures in the medial temporal lobe, improves episodic memory. This is one of very few robustly reproduced cognitive findings in noninvasive brain stimulation. The comprehensive review and detailed meta-analysis by Goicoechea et al. makes a convincing case for efficacy in healthy people and will be important for neuroscientists and clinical researchers in memory and dementia.

In 2014, Wang et al. showed that noninvasive stimulation of a parietal site, connected functionally to the hippocampus, increased resting state functional connectivity throughout a canonical network associated with episodic memory. It also caused a memory boost which was proportional to the connectivity increase within subjects. Their discovery that an imaging biomarker could (1) be used to target a functional network with critical nodes too deep to reach directly with TMS, (2) enable individualized, functionally confirmed, targeting, and (3) provide a scaling measure of target engagement, is one of the signal advances in noninvasive brain stimulation.

The meta-analytical methodology used by these authors is rigorous, and the central finding, viz. that high-frequency, network-targeted stimulation reproducibly improves event recall, is amply supported. The question of whether to stimulate before or after memory encoding is also answered. While there is a hint that individualized anatomical or functional MRI-based targeting may be superior to atlas or group average-based techniques, the finding did not survive correction for multiple comparisons. Additional studies will be needed to resolve this issue, optimize the stimulation delivery parameters, and further define the behavioral effect.

While the authors appropriately emphasize the associated network rather than the hippocampus itself, naming the target after a single node could suggest a primary role for the

hippocampus in the observed outcomes, a conclusion not supported by the data reviewed here. Other nodes in the network are be equally important to aspects of episodic memory and could be useful targets for stimulation.

Despite encouraging results from small clinical samples, the question of efficacy in patients with static lesions and ongoing neurodegeneration remains open. The information gathered here, including the absence of reported adverse events, should spur Phase 2 clinical trials in patients with disorders of memory.

<https://doi.org/10.7554/eLife.108934.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Goicoechea et al. assesses the influence of hippocampal-network targeted TMS to parietal cortex on episodic memory using a meta-analytic approach. This is an important contribution to the literature, as the number of studies using this approach to modulate memory/hippocampal function has clearly increased since the initial publication by Wang et al. 2014. This manuscript makes an important contribution to the literature. In general, the analysis is straightforward and the conclusions are well-supported by the results.

Strengths:

(1) A meta-analysis across published work is used to evaluate the influence of hippocampal-network-targeted TMS in parietal cortex on episodic memory. By pooling results across studies, the meta-analytic effects demonstrate an influence of TMS on memory across the diversity of many details in study design (specific tasks, stimuli, TMS protocols, study populations).

(2) Selectivity with regard to episodic memory vs. non-episodic memory tasks is evaluated directly in the meta-analysis.

(3) The investigation into supplemental factors as predictors of TMS's influence on memory was tested. This is helpful given the diversity of study designs in the literature. This analysis helps to shed light on which study designs, e.g., TMS protocols, etc., are most effective in memory modulation.

Weaknesses:

The authors thoroughly addressed and responded to the prior comments in the revision. The only minor weakness I see is acknowledged in terms of how null effects for particular design or TMS features should be interpreted (i.e., with caution given the regression approach used).

<https://doi.org/10.7554/eLife.108934.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

(1) While the manuscript is written for a scientific audience, the authors are likely aware that findings like this will be of broad appeal to the field of neurology, where treatments for memory loss are desperately needed. For this reason, the authors could consider including a statement regarding an interpretation of this meta-analysis from a clinical standpoint. Statements such as 'safe and effective' imply a clinical indication, and yet the

manuscript does not engage with clinical trials terminology such as blinding, parallel arm versus crossover design, and trial phase. While the authors might prefer not to engage with this terminology, it can be confusing when studies delivering intervention-like five days of consecutive TMS (e.g., Wang et al., 2014) are clustered with studies that delivered online rhythmic TMS, which tests target engagement (e.g., Hermiller et al., 2020). While the 'sessions' variable somewhat addresses the basic-science versus intervention-like approach, adding an explicit statement regarding this in the discussion might help the reader navigate the broad scope of approaches that are utilized in the meta-analysis.

We appreciate the suggestion to enhance interpretability of our report by broader audiences. First, to avoid confusion, we have eliminated “safe” and “effective” descriptors from the main summary of findings in the Abstract (pg. 1) and Discussion (pg. 6). Second, we now describe that reviewed studies included those categorized as traditional clinical trials, as well as non-clinical studies that generally follow clinical trial designs (i.e., multi-day intervention-like studies), in addition to more basic-oriented studies that are geared towards target engagement (Introduction, pg. 2). Third, we now clarify that the Design and Control factors (Figure 3) correspond to fairly standard distinctions in the clinical trials literature and were intended to capture major study design factors choices that are used in both clinical-trial and non-trial studies (Methods, pg. 9; Table S1). Finally, we now clarify that future clinical trials would be needed to evaluate HITS for any specific indication, and that our findings motivate such investigations but do not conclusively indicate efficacy for any given indication (Abstract, pg. 1; Discussion, pg. 7).

Reviewer #1 (Recommendations for the authors):

(1) The color scheme of Figure 1 was a bit confusing. All of the colors used for the flagged regions were incredibly similar. At first glance, it looks like the hippocampus was targeted directly due to the subtle color difference. Could the authors use colors that are more different? Similarly, zooming into the specific locations shows blue dots encompassed by teal. I am not sure what I am looking at here.

We have updated the figure for clarity.

(2) Given the broad appeal of the current study, I would encourage the authors to include a brief visual depiction of “HITS.” This could help the more casual reader to understand the general approach.

We have included this in Figure 1A.

Reviewer #2 (Public review):

(1) While the introduction centers on the role of the hippocampus in episodic memory and posits hippocampal neuromodulation by TMS as causative, the true mechanism may be more complex. Clean hippocampal lesions in primates cause focal loss of spatial and place memory, and I am aware of no specific evidence that the hippocampus does more than this in humans. Moreover, there is evidence that lateral parietal TMS also reaches neighboring temporal lobe regions, which contribute to episodic memory. The hippocampus may, therefore, be a reliable deep seed for connectivity-based targeting of the episodic memory network, but might not be the true or only functional target.

We regret to have implied that we think the hippocampus is the true or only functional target. We agree with the reviewer that the hippocampus is “a reliable deep seed for connectivity-based targeting of the episodic memory network” and that the specific locus/loci of the HITS effects and mechanisms are not yet clear. We now emphasize that although hippocampus is used to define the targeted network, effects of TMS are likely distributed

throughout the network, citing relevant studies that have shown that brain activity changes due to HITS are certainly not restricted to the hippocampus (Introduction, pg. 2).

(2) The meta-analysis combines studies with confirmation of targeting and target-network engagement from fMRI and studies without independent evidence of having stimulated the putative target (e.g., Koch et al). That seems like a more important methodological distinction than merely the use of any individual targeting method. In my experience, atlas-based estimates are at least as accurate as eyeballing cortical areas in individuals. Hence, entering individual functional targeting as a factor might reveal an effect on efficacy.

Our current definition of the “Targeting” factor appears to satisfy this concern. That is, we distinguish studies that used “individual functional targeting” (i.e., resting-state fMRI or DTI connectivity in each individual to select the target) from those that did not (i.e., atlas or other group-average approach). Notably, the Targeting factor modulation effect failed to survive correction for multiple comparisons. We think this satisfies the reviewer criticism, unless the reviewer is suggesting that we categorize studies based on whether they included evaluation of target engagement (e.g., tested for change in fMRI activity or connectivity of the network due to HITS) versus those that measured only behavioral outcomes. We did not include this distinction as a factor, as our analysis focuses on behavioral effects of HITS, and it is not clear what the neural effects would have been in studies in which they were not measured. Notably, we are providing the full raw dataset of effect sizes in a public repository with our final version of record, such that any other categorization schemes could be assessed by others.

(3) The funnel plot and Egger's regression for episodic memory outcomes suggested possible bias, and the average sample size of 23 is small, contributing to the likelihood of false positive results. It would be informative, therefore, to know how many or which studies had formal power estimates and what the predicted effect sizes were.

Regarding the average sample size of 23, we note that we used Hedges' g for the effect size measure because it corrects for bias associated with small samples (pg. 10). Further, small sample sizes contribute to noisy estimates of true effects, allowing outliers to contribute to false positives and low power to contribute to false negatives, but without any reason to systematically yield bias towards false positives. Regarding potential publication bias, although we cannot rule this out based only on the statistics, we think that bias against publication of negative results is unlikely. First, HITS experiments are time consuming and expensive, and most in the field seem to be motivated to publish, whatever the outcome. Second, the notion of memory enhancement via brain stimulation is controversial, and groups have certainly been motivated, if not overly eager, to publish “failure to replicate” studies for HITS (e.g., the failure-to-replicate publication by Hendrikse et al. 2020, which was then re-analyzed by many of the original authors to arrive at different conclusions in Cash et al. 2022). Given these considerations, we think that it is very unlikely that publication bias had any major impact on our conclusions, but of course it cannot be conclusively excluded. Finally, we note that our finding of HITS selectivity for recollection enhancement is likely not affected by publication bias, as this selectivity versus other memory and non-memory outcomes was found only within published studies (i.e., it is very unlikely that publication bias would have led researchers to withhold publication of studies that found effects of HITS on recognition but not on recollection).

(4) In the Discussion, the authors might provide a comparison between the effect size for memory improvement found here with those reported for other brain-targeted interventions and behavioral strategies. It may also be worthwhile pointing out that HITS/memory is one of the very few, or perhaps the only, neuromodulatory effects on cognition that has been extensively reproduced and survived rigorous meta-analysis.

We now emphasize that this is, to our knowledge, the only neuromodulatory effect on cognition that is selective, has been extensively reproduced, and survived rigorous meta-analysis (Discussion, pg. 6). However, we wish to avoid the clinical overinterpretation of our findings that might result if we were to compare directly to effect size estimates for other current therapies, which have been evaluated for specific clinical indications. For example, antibody and pharmacological interventions for Alzheimer's dementia typically have been associated with similar effect sizes to our estimate for HITS. However, those estimates derive from systematic review of randomized controlled trials measuring clinically relevant outcomes at relatively long delays, whereas the HITS studies we review include a mix of controlled and uncontrolled trials, vary in whether clinical outcomes were assessed, and mostly assessed outcomes at shorter delays. Thus, it could be misleading to directly compare the effect sizes. We instead continue to highlight that the HITS effects are promising and warrant rigorous testing for any given clinical indication.

(5) The section of the Discussion on specificity compares HITS to transcranial electrical stimulation without specifying an anatomical target or intended outcome. A better contrast might be the enormous variety of cognitive and emotional effects claimed for TMS of the dorsolateral prefrontal cortex.

We now also note that TMS of lateral frontal cortex has not been associated with similarly high specificity (Discussion, pg. 6). Note however that we cannot exclude anti-depressant or other psychological effects of HITS, as such outcomes were not consistently assessed in HITS studies and so were not included in our analyses.

(6) With reference to why other nodes in the episodic memory network have not been tested, current flow modeling shows TMS of the medial prefrontal cortex is unlikely to be achievable without stronger stimulation of the convexity under the coil, in addition to being uncomfortable. The lateral temporal lobe has been stimulated without undue discomfort.

We now additionally indicate that medial prefrontal stimulation may be ineffective given conventional TMS (Discussion, pg. 7). However, we are aware of no studies that have stimulated the portion of middle temporal gyrus that shows strong connectivity with hippocampus. We have tried this location, which positions the coil on or slightly above the ear and bordering on the temple area that is very sensitive to most. We were not able to minimize pain/discomfort for most subjects in pilot experiments, and so had to abandon it. Perhaps others have succeeded? If the reviewer has any specific references that could be included we would be happy to add them and update this section accordingly.

(7) Finally, a critical question hanging over the clinical applicability of HITS and other neuromodulation techniques is how well they will work on a damaged substrate. Functional and/or anatomical imaging might answer this question and help screen for likely responders. The authors' opinion on this would be informative.

We appreciate this point but don't think there are enough data to assess the level of substrate damage needed to frustrate any stimulation benefits. The only thing we can say is that HITS was equally effective for mild to moderate Alzheimer's dementia as it was for other non-neurodegenerative groups (nonsignificant effect of the Population factor, Figure 3B), suggesting that whatever degree of damage present in that group is insufficient to prevent the stimulation effects. We now highlight this point and raise the issue that, presumably, some level of damage would render HITS ineffective (Discussion, pg. 8).

Reviewer #3 (Public review):

(1) My only significant concern is how studies are categorized in the 'Timing' factor (when stimulation is applied). Currently, protocols in which TMS is administered across days are

categorized as 'pre-encoding' in the Timing factor. This has the potential to be misleading and may lead to inaccurate conclusions. When TMS is administered across multiple days, followed by memory encoding and retrieval (often on a subsequent day), it is not possible to attribute the influence of TMS to a specific memory phase (i.e., encoding or retrieval) per se. Thus, labeling multi-day TMS studies as 'pre-encoding' may be misleading to readers, as it may imply that the influence of TMS is due to modulation of encoding mechanisms per se, which cannot be concluded. For example, multi-day TMS protocols could be labeled as 'pre-retrieval' and be similarly accurate. This approach also pools results from TMS protocols with temporal specificity (i.e., those applied immediately during encoding and not on board during memory testing) and without temporal specificity (i.e., the case of multi-day TMS) regarding TMS timing. Given the variety of paradigms employed in the literature, and to maximize the utility/accuracy of this analysis, one suggestion is to modify the categories within the Timing factor, e.g., using labels like 'Temporally-Specific' and 'Temporally Non-specific'. The 'Temporally-Specific' category could be subdivided based on the specific memory process affected: 'encoding', 'retrieval', or 'consolidation' (if possible). I think this would improve the accuracy of the approach and help to reach more meaningful conclusions, given the variety of protocols employed in the literature.

We agree in principle with this criticism and think that the most straightforward way to address it is to relabel the “Pre-Encoding” category as “Pre-Task”. The issue with labeling/considering single-session stimulation delivered immediately before encoding as “Pre-encoding” is that this makes the assumption that this stimulation doesn’t also affect retrieval (i.e., is temporally specific). We do not have certainty about the timecourse of how a single session of stimulation affects brain activity. We think the “Pre-Task” label and interpretation is the best way to address this, to avoid suggesting that we are confident about the timecourse/selectivity of stimulation effects. Notably, the “Sessions” factor directly compares among designs that delivered stimulation in a single session versus in multiple consecutive sessions, and was a nonsignificant modulator. Thus, our analyses already compare studies that are relatively temporally specific versus those that, likely, are less so. In addition to relabeling, we have also added clear caveats to address the interpretive constraint imposed by the unknown timecourse of stimulation effects (Discussion, pg. 6-7) and revised the Abstract to reflect this change.

(2) As the scope of the meta-analysis is limited to TMS applied to parietal or superior occipital cortex, it is important to highlight this in the Introduction/Abstract. The 'HITS' terminology suggests a general approach that would not necessarily be restricted to parietal/nearby cortical sites.

This was previously highlighted only in the Methods and Discussion (with a Discussion paragraph dedicated to the issue of target selection; see also Comment 6 from Reviewer 2). We now also note this in the Introduction (pg. 2) and Abstract.

Minor:

(1) To reduce the number of study factors tested, data reduction was performed via Lasso regression to remove factors that were not unique predictors of the influence of TMS on memory. This approach is reasonable; however, one limitation is that factors strongly correlated with others (and predict less unique variance) will be dropped. This may result in a misrepresentation, i.e., if readers interpret factors left out of this analysis as not being strongly related to the influence of TMS on memory. I do see and appreciate the paragraph in the Discussion which appropriately addresses this issue. However, it may be worth also considering an alternative analysis approach, if the authors have not already done so, which explicitly captures the correlation structure in the data (i.e., shown in Figure S2) using a tool like PCA or an appropriate factor analysis. Then, this shared covariance amongst factors can be tested as predictors of the influence of TMS -

e.g., by testing whether component scores for dominant PCs are indeed predictive of the influence of TMS. This complementary approach would capture rather than obfuscate the extent to which different factors are correlated and assess their joint (rather than independent) influence on memory, potentially resulting in more descriptive conclusions. For example, TMS intensity and protocol may jointly influence memory.

We argue that feature selection via Lasso regression is a better approach for our research question than PCA, factor analysis, or other latent variable methods. The main reason is that PCA would sacrifice the interpretability of our findings with respect to the design of future experiments using or testing HITS. That is, because PCA creates composite components that are linear combinations of multiple variables, we would lose the ability to provide clear, actionable guidance to researchers about which specific study design choices (e.g., stimulation intensity, protocol type, timing) influence memory outcomes. Given that a major goal of our meta-analysis is to inform future experimental design, we believe that it is essential to maintain interpretability of the individual factors that must be decided when designing a study. Regarding factor analysis, this approach would require making *a priori* theoretical decisions about how to group individual moderators, which could introduce subjective bias into the analysis and would introduce other complications such as a need for validation of the resulting factor scores. We believe that the exploratory nature of our investigation, examining which among many possible study design factors substantially determine TMS efficacy, is better suited to a data-driven selection approach like Lasso. While the reviewer correctly notes that Lasso may drop factors that are correlated with stronger predictors, this feature can be considered advantageous in terms of identifying factors for inclusion in future study designs. That is, this can help identify the most parsimonious set of independent predictors, such that researchers can focus on the study design elements that matter most when controlling for other factors. Notably, we provide the table of factor relationships (Figure S2) so that interested readers can inspect how dropped factors were related to those that were retained.

It is also important to note that we have provided the full dataset with our resubmission, which has been deposited in Dryad with a link in the Data Availability section (pg. 15). Thus, others are free to explore alternative analytical approaches should they wish to examine the data from different perspectives or to answer different questions.

(2) Given the specific focus on TMS applied to parietal cortex to modulate hippocampal and related network function, it would be fruitful if the authors could consider adding discussion/speculation regarding whether this approach may be effectively broadened using other stimulation methods (e.g., tACS, tDCS), how it may compare to other non-invasive brain stimulation methods with depth penetration to target hippocampal function directly (transcranial temporal interference, or transcranial focused ultrasound), and/or how or whether other stimulation sites may or may not be effective.

We briefly discuss a meta-analysis of tACS studies which reported nonspecific effects, including for parietal targets overlapping those used for HITS (Discussion, pg 6). We briefly speculate about how tES effects remain mechanistically uncertain. We are afraid that further speculation about other stimulation modalities and targets would be beyond the scope of this focused meta-analysis, given especially the few datapoints for newer approaches such as TI or tFUS.

(3) Studies were only included in the meta-analysis if they contained objective episodic memory tests. How were studies handled that included both objective and subjective memory, or other non-episodic memory measures? For example, Yazar et al. 2014 showed no influence of TMS on objective recall, but an impairment in subjective confidence. I assume confidence was not included in the meta-analysis. Similarly, Webler et al. 2024 report results from both the mnemonic similarity task (presumably included)

and a fear conditioning paradigm (presumably excluded). Please clarify in the methods how these distinctions were handled.

Studies were included in our meta-analysis if they included at least one objectively scorable test of episodic memory. We only included objectively scorable test performance in our analysis, excluding scores from any other subjective measures if they were also reported. This is now clarified in Methods (pg. 9).

(4) The analysis comparing memory to non-memory measures is important, showing the specificity of stimulation. Did the authors consider further categorizing the non-memory tasks into distinct domains (i.e., language, working memory, etc.)? If possible, this could provide a finer detail regarding the selectivity of influences on memory vs. other aspects of cognition. It is likely that other aspects of cognition dependent on hippocampal function may be modulated as well, i.e., tasks with high relational/associative processing demands.

This is an interesting idea, but it is beyond our expertise to categorize these other tasks based on the nature of processing demands that they capture. Note that the task names are provided in the data table that we are making available online with our submission of record (via Dryad), such that other groups could address this question if interested.

(5) In the analysis of the Intensity factor, how were studies using Active (rather than resting) MT categorized? Only resting MT is mentioned in Table S1. This is important as the original theta-burst TMS protocol from Huang et al. 2005 determines intensity based on Active Motor Threshold.

MT was resting/passive in all reviewed studies except for one (Tambini et al. 2018), which used 80% of active MT. We categorized this as <100% MT for the Intensity factor, as it was <100% of MT as defined in that study. Although one could make the argument that 80% AMT might instead correspond to 100+% RMT, this change would have very little influence on our results or conclusions. We now clarify this in Table S1.

(6) Is there a reason why the study by Koen et al. 2018 (Cognitive Neuroscience) was not included? TMS was performed during encoding to the left AG, and objective memory was assessed, so it would seemingly meet the inclusion criterion.

The failure to include Koen et al. 2018 was our error. Koen et al. 2018 is the only study that used “online” stimulation, delivered during the trials when memoranda were displayed for encoding in the task. In contrast, all other reviewed studies delivered “offline” stimulation either before the memoranda was presented (“Pre-Task”) or after the encoding period but before retrieval (“Post-Encoding”). Therefore, categorization for the “Timing” factor would be problematic for its inclusion in the main analysis. We therefore now include Koen et al. 2018 in the “Supplementary Results” section as well as the corresponding main Results section on “Similar outcomes in studies that were excluded from meta-analysis”. We also note in the relevant discussion that “online” stimulation, as done in Koen et al. 2018, is typically considered disruptive (e.g., Beynel et al. 2019 Neuroscience & Biobehavioral Reviews; Yeh & Rose 2019 Frontiers in Psychology), which should be taken into account when considering the findings of Koen et al. 2018 relative to other reviewed studies that used “offline” designs.

(7) It would be helpful to briefly differentiate the current meta-analysis from that performed by Yeh & Rose (How can transcranial magnetic stimulation be used to modulate episodic memory?: A systematic review and meta-analysis, 2019, Frontiers in Psychology) (other than being more current).

Beyond being more current and therefore including many more studies in which stimulation targets were based on hippocampal connectivity (which tend to have been published more recently), the differences with Yeh & Rose 2019 are subtle. Our review focuses on assessment

of network targeting and whether effects were specific to episodic memory versus other tasks, which differs somewhat from the focus of Yeh & Rose 2019. The main difference in conclusions likely derives from there being more network-focused memory TMS experiments now than were available for Yeh & Rose's review. We also differentiate episodic memory into recollection versus other components to test specificity and analyze modulation by many study design factors relevant to HITS studies that were not emphasized in Yeh & Rose's review. Note that we now cite Yeh & Rose for those interested in potential differences.

(8) For transparency and to facilitate further understanding of the literature and potential data re-use, it would be great if the authors consider sharing a supplementary table or file that describes how individual studies/memory measures were categorized under the factors listed in Table S1.

As promised in our original submission, we are providing the full data table, including how individual studies and memory measures were categorized, as an open dataset in Dryad. The Dryad dataset is cited in "Data availability" (pg. 15).

Reviewer #3 (Recommendations for the authors):

Please explicitly state in the Methods (Meta-analysis of effect modifiers section) that the criteria used for categorizing each measure into a factor (e.g., probing Recollection, Recognition, etc.) are fully described in Table S1; this will help readers to find these details (it took me a while!).

This is now emphasized (pg. 10).

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